

Brief Project Description

Submission-ready summary | Snapdragon AI Lab Build & Present Challenge 2026

PROJECT TITLE

BuildMesh: Local-First Construction Intelligence

WHAT BUILDMESH DOES

BuildMesh is a local-first construction intelligence platform that connects site observations, worker updates, planned quantities, weather context, tasks, and decisions in a persistent project graph. Its OpenMesh-style agent runtime turns these signals into evidence-backed recommendations while keeping consequential actions under human approval.

MVP WORKFLOW

Observe -> Reconcile -> Assess risk -> Recommend -> Human approves -> Act + remember

Example: a site worker records Foundation F-12 as 85% complete with 35.5 m3 placed against a 42 m3 plan. BuildMesh records the update as evidence, derives progress, checks consistency, joins a sourced rain forecast, and creates a reviewable recommendation to validate weather-sensitive work. The follow-up task is created only after a named site engineer approves it.

TECHNICAL IMPLEMENTATION

The current backend is a Python/FastAPI service with SQLite-backed graph storage. It records projects, nodes, relationships, evidence, events, agent-run hashes, recommendations, reviews, and approved tasks. Bounded Project State, Risk, and Recommendation agents are orchestrated through OpenMesh-style workflow logic. Automated tests cover the review gate and progress-reconciliation conflict path.

SNAPDRAGON RELEVANCE AND RESPONSIBLE DEPLOYMENT

BuildMesh is designed for Snapdragon-powered PCs: sensitive site images, local project memory, document extraction, and agent orchestration can remain on-device. A QNN/ONNX provider boundary is included for a Qualcomm AI Hub or open-source model runner. Weather, traffic, and maps remain explicitly sourced external context. The project does not claim target-device latency or model performance before that deployment is benchmarked.

Why it matters

BuildMesh makes fragmented construction evidence legible, traceable, and actionable without replacing professional engineering judgment.